Price Predictor

Predicting Listing Prices Across NYC, LA & Chicago

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What We'll Cover

1. Objective & Value Proposition
2. Data Sources
3. EDA Highlights
4. Modeling Approach
5. Results & Insights
6. Challenges & Future Work

Objective & Value Proposition

The Problem

- Airbnb hosts struggle to price listings competitively across different cities and neighborhoods
- Without data, they're guessing

Our Goal

- Build a multi-city price prediction model using listing features, location data, and walkability scores
- Focused on 3 major US cities: NYC, LA, and Chicago

Why it Matters

- Accurate pricing helps hosts maximize revenue and helps guests find fair value
- Better outcomes for both sides of the marketplace

62,544 listings analyzed across 3 cities

Data Sources

Inside Airbnb

- Free public CSV listings data for NYC, LA, and Chicago
- Includes price, room type, amenities, location, reviews, host attributes, & more

Inside Airbnb
Adding data to the debate

Walk Score API

- Walkability, transit, and bike scores fetched per listing location
- Free tier limits of 5,000 requests/day

Walk Score®

Combined into one unified Parquet dataset using Polars

Data Pipeline

Collect

- Pandas
- Download CSVs from Inside Airbnb
- Fetch Walk Scores via API
- Fetching with requests caching, logging, caffeinate, flushing, etc.

Clean

- Pandas
- IQR-based outlier capping
- Median imputation for missing values
- Parse amenities from raw text strings

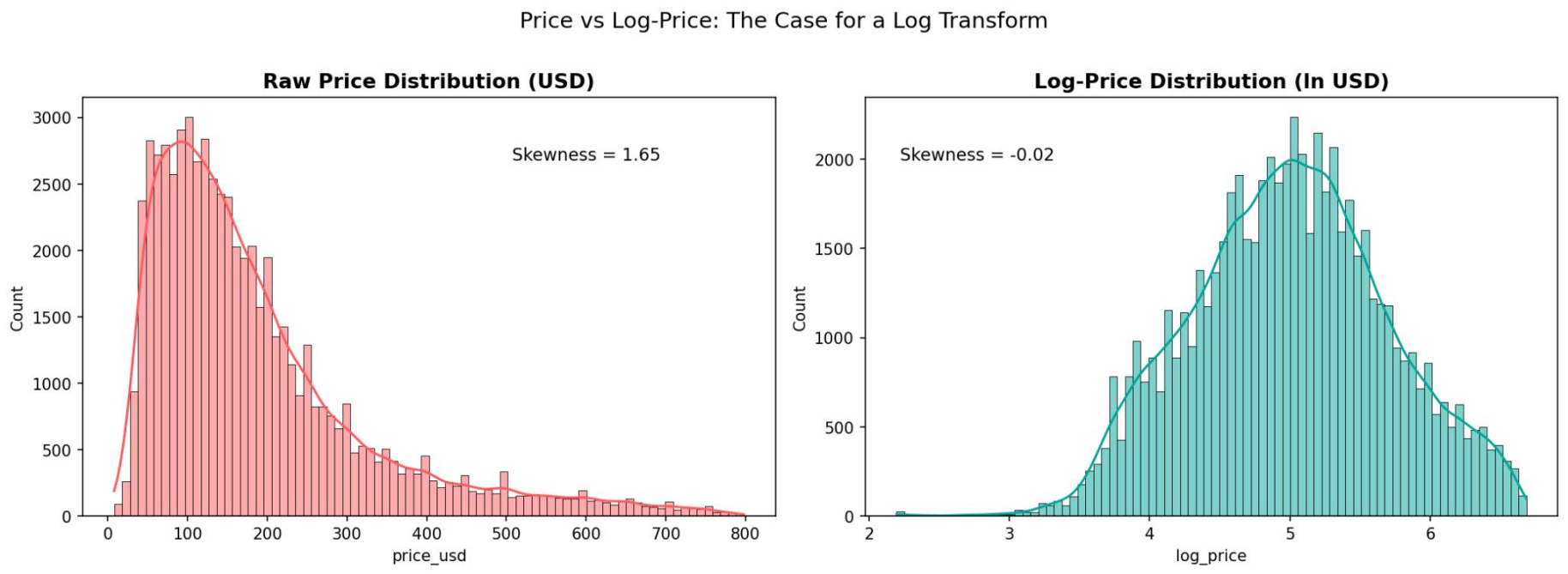
Engineer

- Polars
- Interaction terms
- Binary amenity flags
- Categorical encoding for room type, city, and neighborhood
- Variable correlations

Model

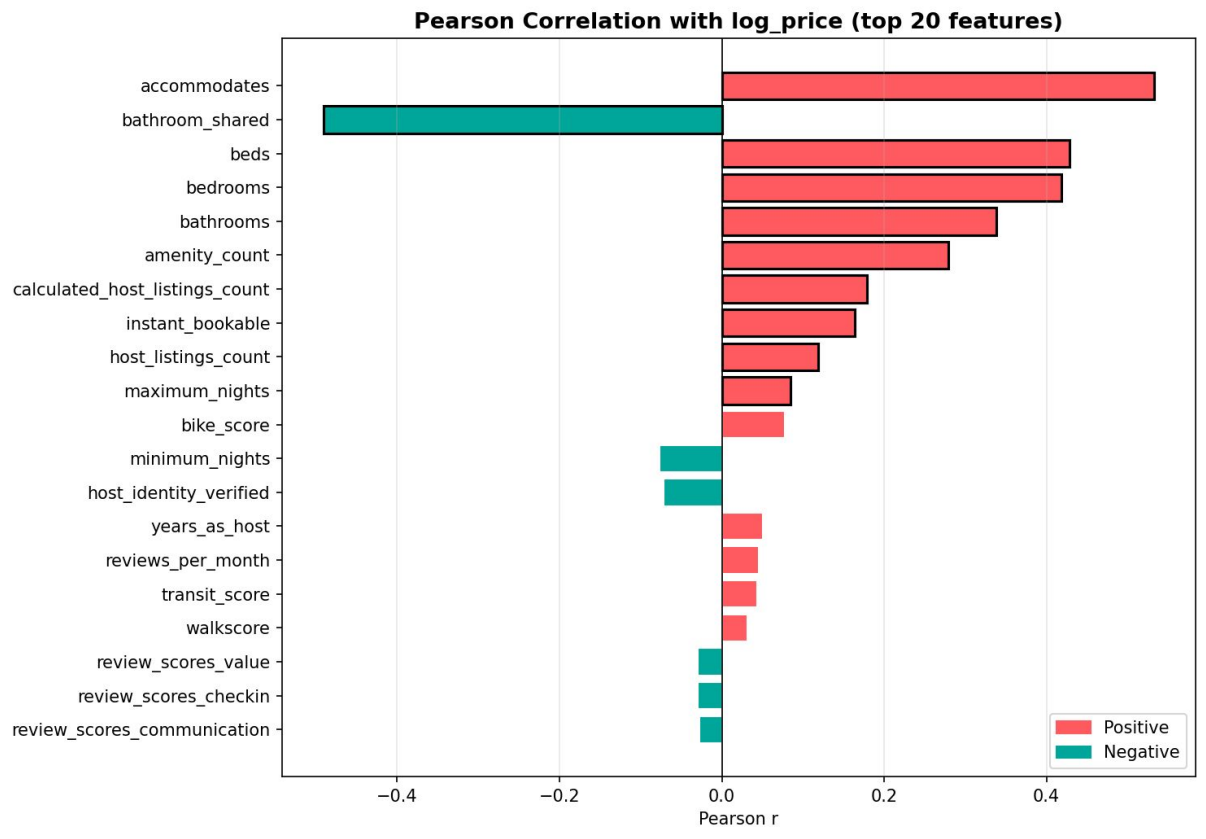
- Train/test split (80-20)
- RidgeCV (LassoCV, Elastinet)
- Random Forest (GridSearch CV)
- XGBoost (Bayesian hyperparameter tuning)
- Model explanability with SHAP
- Feature importance (coefficients, permutation)
- Residual analysis

EDA: Price Distributions



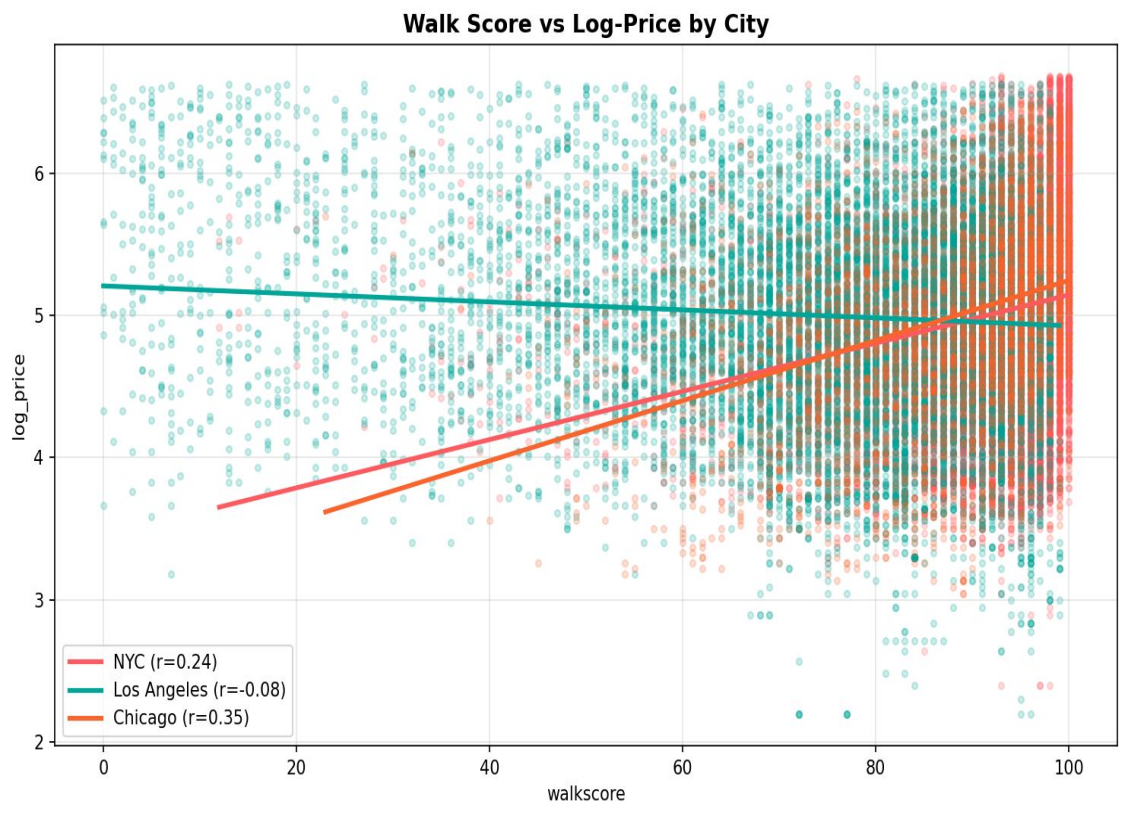
- Raw Airbnb prices are **heavily right-skewed** (skewness = 1.65), meaning most listings cluster under $200 but a long tail of luxury properties drags the mean upward (a classic problem for regression models)
- Taking the **natural log of price nearly perfectly normalizes the distribution**, bringing skewness down to just -0.02

EDA: Features



- Capacity dominates: **accommodates is the strongest single predictor** ($r \approx 0.55$), followed closely by beds, bedrooms, and bathroom
- Tells you that listings are priced primarily around **how many people they can host**
- Location/host quality features like walkscore, transit_score, and host_identity_verified have notably weaker correlations

EDA: Walkscore



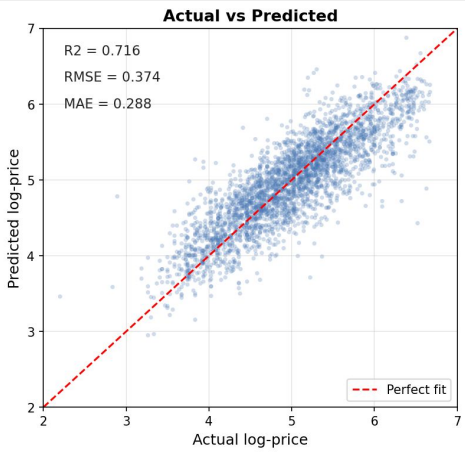
- The relationship between **walkability and price is surprisingly inconsistent** across cities
- **Walkability matters most in Chicago and NYC** (both walkable, transit-oriented cities)
- **LA is nearly flat** (r = 0.08)
  - LA is a car-dependent city where walkability just isn't how neighborhoods are valued
  - Walkscore carries little pricing signal there

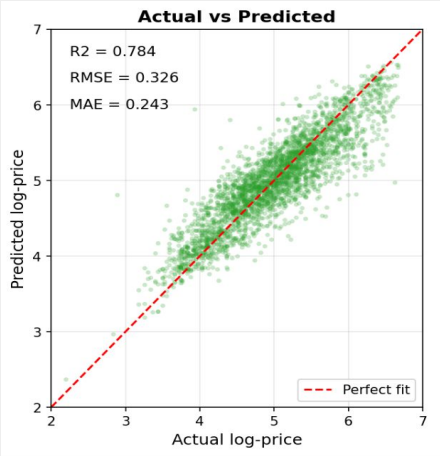
Modeling Approach

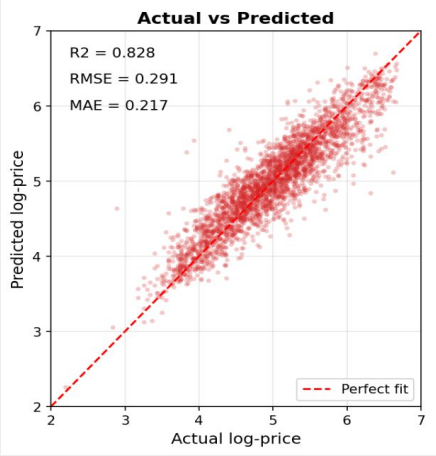
Ridge Regression	A linear model that predicts log-price as a weighted sum of all features, with L2 regularization that shrinks large coefficients to prevent overfitting: making it stable even with 200+ correlated amenity features
Random Forest	An ensemble of 50+ decision trees, each trained on a random subset of data and features, whose predictions are averaged together, reducing variance and capturing non-linear patterns Ridge misses
XGBoost	A gradient boosting model that builds trees sequentially, with each new tree correcting the errors of the previous one, making it the most powerful of the three at capturing subtle, non-linear price patterns

Model Results

Model	R^2	RMSE	Notes
Ridge Regression	0.73	0.36	Baseline: interpretable, stable
Random Forest	0.78	0.32	Strong feature importance signal
XGBoost	**0.82**	**0.29**	Best performer: Bayesian tuned







Insights, Challenges, & Future Work

Implications

- Pricing is **largely driven by physical capacity** (accommodates, bedrooms, bathrooms) and room type across all 3 models, meaning hosts have direct, **actionable levers to optimize** their listing price without relying on location alone
- Performance gap between Ridge and XGBoost confirms that **Airbnb pricing is inherently non-linear**

Challenges

- Walk Score **API rate limits** (5,000 req/day)
  - Incomplete coverage -> NaNs
- Price **outliers varied significantly** by city: IQR thresholds had to be calibrated per-city rather than globally
- Amenity parsing from raw text strings introduced noise; **standardizing amenity names required manual cleaning**

Future Work

- Incorporate **seasonal and event-driven pricing signals** from calendar availability data
- Add **review sentiment scores** using NLP on guest review text
- Expand to additional cities and build a **real-time pricing recommendation API**

Thank you!